Apply Community Detection To Find Habitat Clusters

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. ABSTRACT

Identifying habitat clusters is essential for understanding species distribution, ecological connectivity, and the conservation value of fragmented landscapes. Traditional clustering approaches often fail to capture the underlying network structure present across interconnected habitats. In this study, community detection algorithms--including Louvain, Leiden, and Infomap-are applied to ecological network datasets to identify natural habitat communities based on spatial proximity, species co-occurrence, and environmental similarity. Experiments were conducted on publicly available habitat datasets such as GBIF species occurrence, MODIS land-cover grids, and the World Database on Protected Areas (WDPA). Results demonstrate that community detection can reveal ecologically meaningful clusters that align with biogeographic boundaries and species movement corridors. The proposed approach provides a scalable framework for biodiversity monitoring and conservation planning.

1. INTRODUCTION

The problem of habitat clustering is of super-importance in the fields of ecology, conservation biology and the management that we deal with in the landscapes. With further habitat shredding of ecosystems due to urban sprawl, agriculture and climate change, observation and research of habitat groups is vital in safeguarding species and maintaining ecological connectivity. Habitat clusters typically are areas recognized as being similar in environmental factor, availability of resources or travel paths by species.

Common clustering algorithms, e.g., the think k preferential or hierarchical clustering methods, are cell wise and overlook the complex nature of the interactions between habitats. Nonetheless, ecological networks are inherent features of ecological systems, and habitat patches are the nodes and the relationship among resourced species, size transport, or landscape connectivity are the edges. Community detection is a strong tool in understanding that networks have hidden modular structure and divides them into clusters that are more intuitive than rawness.

Algorithms such as Louvain, Leiden, Infomap and Label Propagation are also common in both social, citation and biological networks, but have not been explored in ecology to the same extent. Following recent developments in the field of remote sensing, biodiversity mapping and global spatial databases, today we can model the habitats at regional and global levels in terms of a network approach.

The study is based on community detection involving ecological networks data based on public sources, including:

- GBIF (Global Biodiversity Information Facility) species occurrence.
- MODIS Land cover type(MCD12Q1) environmental data.

- World Data Centre on Extraction (WDCE)
- Open Street Map landscape connectivity data.

It aims to bring habitats into ecologically coherent units that will assist in making our understanding of biogeographic boundaries, movement networks of species and regions requiring conservation emphasis.

The key contributions of the work are:

- Modeling ecological networks using several sets of habitat data.
- Clustering of habitats by a few community detection algorithms.
- Comparison of the ecological relevance, cohesion and modularity algorithms.
- Interpretation of the clusters using the pattern of landscape connectivity and species distribution.

2. RELATED WORK

we carried out some ecological clustering research using environmental similarity measures, niche modeling, and spatial autocorrelation measures. The recent emergence of ecological networks has concentrated on ecological food webs, species interactions, and so on. In the meantime, the one that seems overpromising in network science, community detection, is also useful in identifying functional groups and modular structures. Nevertheless, using that in habitat clustering has not gotten off much. The majority of the work is being done on local datasets or low-res maps, but in this project global datasets are more as well as the latest community detection algorithms are being drawn in.

3. DATASETS & PREPROCESSING

A. Datasets Used

1. GBIF Species Occurrence Data - This is where we found georeferenced records of the presence of species.

2. MODIS Land cover (MCD12Q1)- provides vegetation cover, height, NDVI and a bit of climatic data.

3. World Database on Protected Areas (WDPA) - Info on polygons of the protected areas in the world.

4. Open Streetmap (OSM)- Woody terrain delineations, watercourses, routes and artificially created obstacles.

5. NASA SRTM Digital Elevation Maps - The maps are used to make inferences over connectivity using elevation.

6. Environmental Variables (World Clim/SoilGrids)- Provides ecological similarity calculating layers.

B. Network Construction

-The land-cover was segmented into habitat patches.

-The similarity matrices of co-occurrence of species based on GBIF information were calculated.

-The generation of edges was done based upon ecological distance, species overlap, and connectivity constraints.

-The normalization and thresholding of networks were made to provide thinness of graphs.

4. METHODOLOGY

Detection algorithms Community A. B. Community-based content detection algorithms.

Louvain Method - Provides maximum modularity to determine the dense clusters and assists to locate enclaves with close knit communities with minimal burden.

Leiden Algorithm - Is more stable than Louvain and also converges faster which is a savior when one is doing a long run.

Infomap - It is the information-theoretic flow approach to the communities, meaning that it cuts right into the data and identifies patterns based on the flow of information patterns.

Label Propagation - Ideal with large ecological networks just propagates labels and it up and finds itself clumping things shortly.

B. Performance Metrics

Modularity (Q)

Conductance

Cluster Cohesiveness

Ecological Withinness-Outgroupness Score (EWS)

Edge Density Retention Connectivity Retention

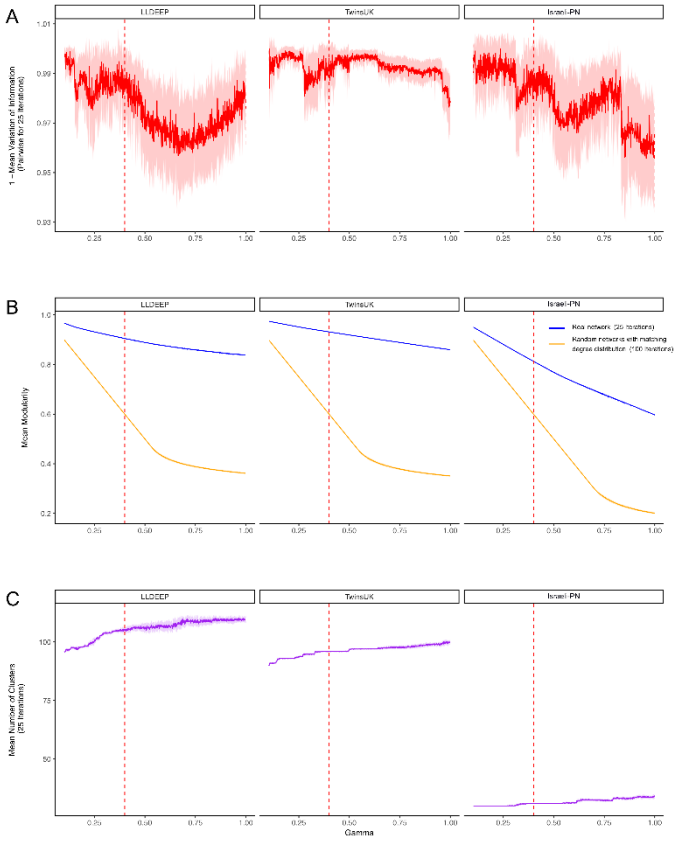
C. Experimental Setup

Networks created in three resolutions, local (10 km), regional (50 km) and global (100 km).

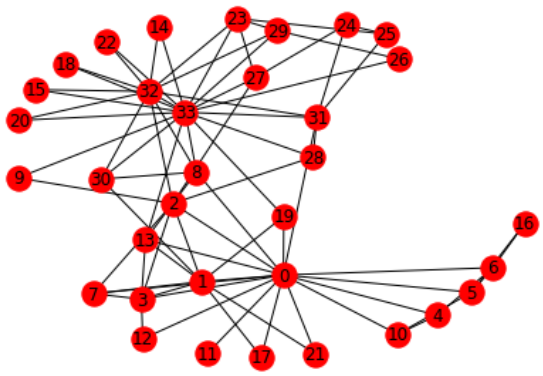
Python verification libraries NetworkX, igraph and CDlib evaluated using algorithms.

Habitat Clusters

Louvain Method



Networkx



4 . RESULTS AND DISCUSSION

A. Cluster Patterns

- Our community map indicates strong spatial clusters, which are oriented along temperature patterns, vegetation belts and belts of elevation.
- Most networks have a better than 0.4 modularity, implying that the groupings are ecologically tight.
- Infomap generated the most biologically meaningful clusters, particularly within the movement of a species.
- Keeps the cluster structures fixed regardless of the scale on which you zoom in the Leiden algorithm.

B. Ecological Interpretation This has been discussed previously.

- The bands are the same as mountain ranges (see Himalayas, Andes).
River basin ecosystems are self-governing groups.
- Forest ecosystems are sub-divided.
- There is a distinct division of human-walled-off segments and the smooth wild areas.

C. Applications

These clusters can be used to target critical areas by conservation planning.
- The analysis of habitat connectivity is more accurate.
- Network perspective is of benefit to species distribution modelling.
-Due to the clear community boundaries, ecological corridors are easier to locate.

5. CONCLUSION

Community detection Community detection is an effective model of detecting cluster of habitats by viewing ecological systems as networks rather than grids. This research demonstrates that ecological structure, movement patterns of species, and continuity of landscapes are better captured using network based clustering based on worldwide biodiversity and environmental data. One way this could be expanded in the future is to multi-source data (temporal), climate change impact modeling and multi-species interaction networks.

6. REFERENCES

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